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Q1:

1: Regression with Time Fixed Effects

$Crop_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_t + \gamma_t + u_{it} = \beta_1 X_{it} + \theta_t + u_{it}$

$(\theta_t = \beta_0 + \beta_2 Z_t + \gamma_t)$

$Crop_{it} - \frac{1}{I}\sum_{i=1}^{I} Crop_{it} = \beta_1 (X_{it} - \frac{1}{I}\sum_{i=1}^{I} X_{it}) + (u_{it} - \frac{1}{I}\sum_{i=1}^{I} u_{it})$

By using "Time-demeaned" OLS regression, researcher could estimate the β_1.

2. Regression with Entity Fixed Effects

$Crop_{it} = \beta_0 + \beta_2 Z_i + \beta_1 X_{it} + \gamma_t + u_{it}$

$\alpha_i = \beta_0 + \beta_2 Z_i$

$Crop_{it} = \alpha_i + \beta_1 X_{it} + \gamma_t + u_{it}$

$Crop_{it} - \frac{1}{T}\sum_{t=1}^{T} Crop_{it} = \beta_1 (X_{it} - \frac{1}{T}\sum_{t=1}^{T} X_{it}) + (\gamma_t - \frac{1}{T}\sum_{t=1}^{T} \gamma_t) + (u_{it} - \frac{1}{T}\sum_{t=1}^{T} u_{it})$

By using "Entity-demeaned" OLS regression, researcher could estimate the β_1.

Q2:

1. $DID = \frac{1}{n}\sum_{i=1}^{n} (Y_i^1 - Y_i^0) \times D_i - \frac{1}{n}\sum_{i=1}^{n} (Y_i^1 - Y_i^0) \times (1 - D_i)$

$= \frac{1}{n}\sum_{i=1}^{n} (Y_i^1 - Y_i^0)(2D_i - 1)$

2. Assume $E[Y_{oi(1)} - Y_{oi(0)} | D_i = 1] = E[Y_{oi(1)} - Y_{oi(0)} | D_i = 0]$

$Y_{oi(1)}$: In Gansu province, but is not effected by the policy

$Y_{oi(0)}$: Out of Gansu province, not effected by the policy

3. The DID estimator may over estimate the treatment effect. Because of the increase of the farmer's production enthusiasm, the Y_i^1 in Gansu province will be higher, but not directly because of the policy. However, the crop in other provinces remain the same. So the DID estimator will be over estimate.

Q3:

1. Because the two states have initial difference. Comparing New Jersey v.s. PA after the wage change may not be the true results that the minimum wage policy takes.

2. $Empl_{it} = \beta_0 + \beta_1 Group_t + \beta_2 Treatment_i + \beta_3 (Group_t \times Treatment_i) + \varepsilon_{it}$

3. $E[Y_{0i}(1) - Y_{0i}(0) \mid Group=1] = E[Y_{0i}(1) - Y_{0i}(0) \mid Group=0]$

$Y_d(t)$. $D=0$, control units
$D=1$, Treated units
$T=0$, pre-treatment period
$T=1$, post-treatment period

4.

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. reg empl group##treatment fries soda hoursopening cowned
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Source	SS	df	MS	Number of obs	=	757
				F(7, 749)	=	10.88
Model	4613.79535	7	659.113622	Prob > F	=	0.0000
Residual	45355.457	749	60.5546822	R-squared	=	0.0923
				Adj R-squared	=	0.0838
Total	49969.2523	756	66.0968946	Root MSE	=	7.7817

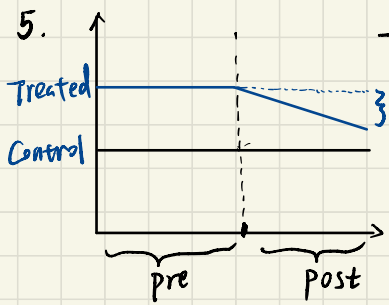
empl	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
1.group	-3.073359	1.055383	-2.91	0.004	-5.145219	-1.001498
1.treatment	-1.470434	1.298612	-1.13	0.258	-4.019787	1.078919
group#treatment						
1 1	4.105866	1.432743	2.87	0.004	1.293197	6.918535
fries	.057069	3.551448	0.02	0.987	-6.914908	7.029045
soda	2.23695	4.216988	0.53	0.596	-6.041572	10.51547
hoursopening	.5110041	.0706844	7.23	0.000	.3722409	.6497673
cowned	-1.088449	.60109	-1.81	0.071	-2.268471	.0915723
_cons	1.252582	3.44593	0.36	0.716	-5.512248	8.017412

$\therefore Empl = \underset{(3.44593)}{1.252582} - \underset{(1.055383)}{3.073359}\, Group - \underset{(1.298612)}{1.470437}\, Treatment$

$+ \underset{(1.432743)}{4.105866}\, Group \times Treatment + \underset{(3.551448)}{.057069}\, fries + \underset{(4.216988)}{2.23695}\, soda$

$+ \underset{(.0706844)}{0.5110041}\, hoursopening - \underset{(.60109)}{1.088449}\, cowned$

5.



The DID helps to control for time-invariant unobservable factors by differencing out the common trends between the treatment and control groups. If the "parallel trends" assumption holds, any difference in employment changes between the two groups after wage change can be attributed to the treatment effect